

Lab sheet-3

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```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
from scipy import stats
```

```
import statsmodels.api as sm
```

```
# Step 1 & 2: Load the Dataset and Import Libraries
```

```
filename = "HR Employee-Attrition.csv"
```

```
df = pd.read_csv(filename)
```

```
print(f"Dataset successfully loaded. Shape: {df.shape}\n")
```

```
# Step 3: Descriptive Statistics for Numerical Features
```

```
numerical_cols = ['Age', 'MonthlyIncome', 'TotalWorkingYears', 'YearsAtCompany']
```

```
# Mean, Median, Variance, Standard Deviation
```

```
desc_stats = df[numerical_cols].agg(['mean', 'median', 'var', 'std']).T
```

```
# Compute Mode
```

```
desc_stats['mode'] = [df[col].mode()[0] for col in numerical_cols]
```

```
print("=" * 60)
```

```
print("DESCRIPTIVE STATISTICS")
```

```
print("=" * 60)
```

```
print(desc_stats.round(2))
```

```
print()
```

```
# Step 4: Pearson Correlation Analysis & Automatic Plot Saving
```

```
corr_matrix = df[numerical_cols].corr(method='pearson')
```

```
print("=" * 60)
```

```
print("PEARSON CORRELATION MATRIX")
```

```
print("=" * 60)
```

```
print(corr_matrix.round(3))
```

```

print()

# Generate and auto-save Heatmap
plt.figure(figsize=(7, 5))

sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f", vmin=-1, vmax=1)

plt.title("Correlation Matrix of Employee Metrics")

plt.tight_layout()

# Automatically save before showing
plt.savefig("correlation_matrix.png", dpi=300, bbox_inches='tight')

print("[INFO] Figure saved automatically as 'correlation_matrix.png'")

# Non-blocking show prevents terminal execution from halting
plt.show(block=False)

# Step 5 & 6: Independent Two-Sample t-Test

# Business Question: Does monthly income differ between employees who stay vs leave?
# H0: Mean MonthlyIncome(Attrition == 'Yes') == Mean MonthlyIncome(Attrition == 'No')
# H1: Mean MonthlyIncome(Attrition == 'Yes') != Mean MonthlyIncome(Attrition == 'No')

income_left = df[df['Attrition'] == 'Yes']['MonthlyIncome']
income_stayed = df[df['Attrition'] == 'No']['MonthlyIncome']

# Welch's t-test (equal_var=False)
t_stat, p_val_ttest = stats.ttest_ind(income_left, income_stayed, equal_var=False)

print("\n" + "=" * 60)

print("HYPOTHESIS TEST 1: TWO-SAMPLE T-TEST")

print("=" * 60)

print(f"Mean Income (Left): ${income_left.mean():.2f}")
print(f"Mean Income (Stayed): ${income_stayed.mean():.2f}")

print(f"t-statistic:      {t_stat:.4f}")
print(f"p-value:           {p_val_ttest:.4e}")

if p_val_ttest < 0.05:
    print("Decision: Reject the Null Hypothesis (H0).")

```

```
print("Conclusion: Significant difference in monthly income between employees who leave  
and stay.")
```

```
else:
```

```
    print("Decision: Fail to reject the Null Hypothesis (H0).")
```

```
print()
```

```
# Step 7: One-Way ANOVA
```

```
# Business Question: Does monthly income vary across different Job Roles?
```

```
# H0: Mean MonthlyIncome is equal across all Job Roles
```

```
# H1: At least one Job Role has a different mean MonthlyIncome
```

```
role_groups = [group['MonthlyIncome'].values for _, group in df.groupby('JobRole')]
```

```
f_stat, p_val_anova = stats.f_oneway(*role_groups)
```

```
print("=" * 60)
```

```
print("HYPOTHESIS TEST 2: ONE-WAY ANOVA")
```

```
print("=" * 60)
```

```
print(f"F-statistic: {f_stat:.4f}")
```

```
print(f"p-value: {p_val_anova:.4e}")
```

```
if p_val_anova < 0.05:
```

```
    print("Decision: Reject the Null Hypothesis (H0).")
```

```
    print("Conclusion: Monthly income differs significantly across job roles.")
```

```
else:
```

```
    print("Decision: Fail to reject the Null Hypothesis (H0).")
```

```
print()
```

```
# Step 8 & 9: Simple Linear Regression (OLS)
```

```
# Dependent (y): MonthlyIncome, Independent (X): TotalWorkingYears
```

```
X = sm.add_constant(df['TotalWorkingYears'])
```

```
y = df['MonthlyIncome']
```

```
model = sm.OLS(y, X).fit()
```

```
print("=" * 60)

print("LINEAR REGRESSION ANALYSIS")

print("=" * 60)

print(model.summary())

# Plot and auto-save Regression Line

plt.figure(figsize=(8, 5))

sns.regplot(

    x='TotalWorkingYears',

    y='MonthlyIncome',

    data=df,

    scatter_kws={'alpha': 0.3},

    line_kws={'color': 'red'}

)

plt.title("Regression: Monthly Income vs Total Working Years")

plt.xlabel("Total Working Years")

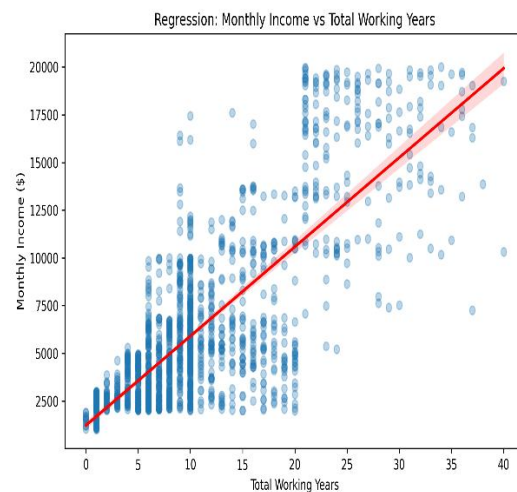
plt.ylabel("Monthly Income ($)")

plt.tight_layout()

plt.savefig("regression_line.png", dpi=300, bbox_inches='tight')

print("[INFO] Figure saved automatically as 'regression_line.png'

plt.show()
```



```
=====
DESCRIPTIVE STATISTICS
=====
| | | | | mean median var std mode
Age 36.92 36.0 83.46 9.14 35
MonthlyIncome 6502.93 4919.0 22164857.07 4707.96 2342
TotalWorkingYears 11.28 10.0 60.54 7.78 10
YearsAtCompany 7.01 5.0 37.53 6.13 5
=====
PEARSON CORRELATION MATRIX
=====
| | Age MonthlyIncome TotalWorkingYears YearsAtCompany
Age 1.000 0.498 0.680 0.311
MonthlyIncome 0.498 1.000 0.773 0.514
TotalWorkingYears 0.680 0.773 1.000 0.628
YearsAtCompany 0.311 0.514 0.628 1.000
=====
[INFO] Figure saved automatically as 'correlation_matrix.png'
=====
HYPOTHESIS TEST 1: TWO-SAMPLE T-TEST
=====
Mean Income (Left): $4787.09
Mean Income (Stayed): $6832.74
t-statistic: -7.4826
p-value: 4.4336e-13
Decision: Reject the Null Hypothesis (H0).
Conclusion: Significant difference in monthly income between employees who leave and stay
=====
HYPOTHESIS TEST 2: ONE-WAY ANOVA
=====
F-statistic: 810.2141
p-value: 0.0000e+00
Decision: Reject the Null Hypothesis (H0).
Conclusion: Monthly income differs significantly across job roles.
=====
LINEAR REGRESSION ANALYSIS
=====
| | | | | OLS Regression Results
=====
Dep. Variable: MonthlyIncome R-squared: 0.597
Model: OLS Adj. R-squared: 0.597
Method: Least Squares F-statistic: 2178.
Date: Sun, 06 Sep 2026 Prob (F-statistic): 2.73e-292
Time: 13:11:09 Log-Likelihood: -13848.
No. Observations: 1470 AIC: 2.770e+04
Df Residuals: 1468 BIC: 2.771e+04
Df Model: 1
Covariance Type: nonrobust
=====
| | | | | coef std err t P>|t| [0.025 0.975]
-----
const 1227.9353 137.299 8.944 0.000 958.612 1497.25
TotalWorkingYears 467.6584 10.021 46.669 0.000 448.002 487.31
=====
Omnibus: 47.473 Durbin-Watson: 1.993
Prob(Omnibus): 0.000 Jarque-Bera (JB): 79.304
Skew: 0.269 Prob(JB): 6.02e-18
Kurtosis: 4.003 Cond. No. 24.2
```